



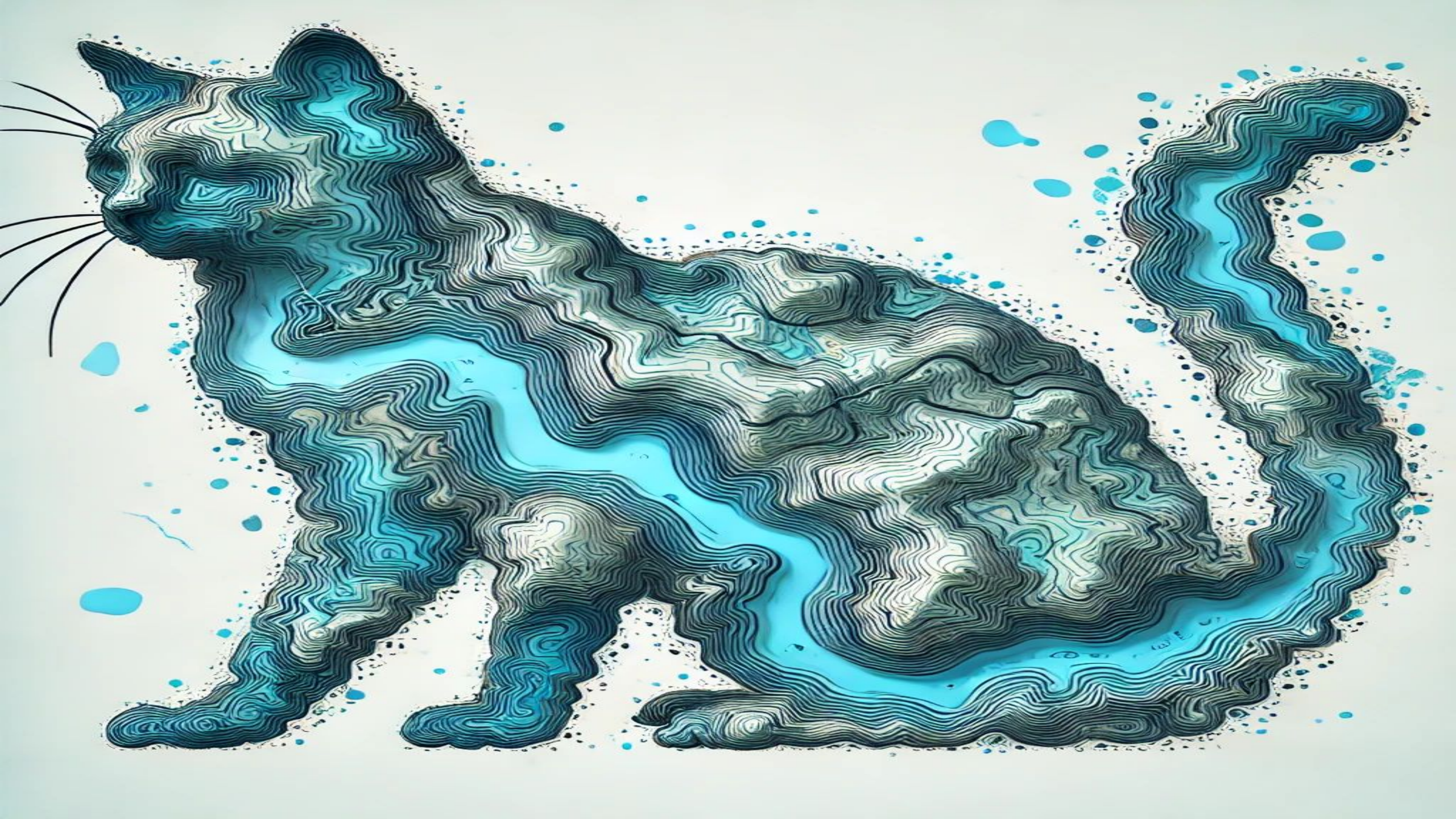
NATIONAL WEATHER SERVICE

Building a Weather-Ready Nation

Assessment of National Weather Service Categorical Flood Inundation Maps (CatFIMs) in Preparation for their Public Release

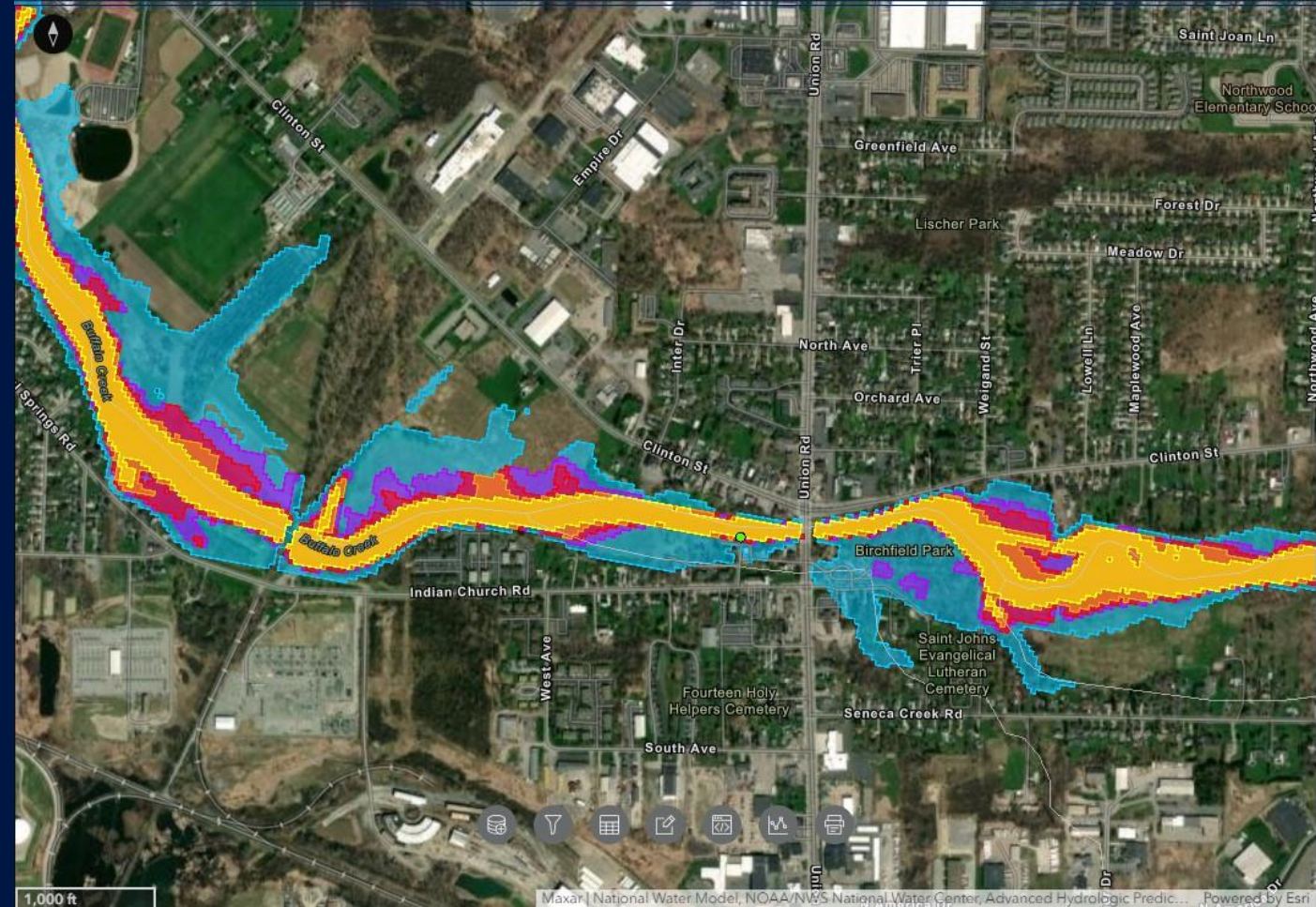
JANUARY 15, 2025

Bill Saunders, NERFC



Categorical Flood Inundation Maps (CatFIMs)

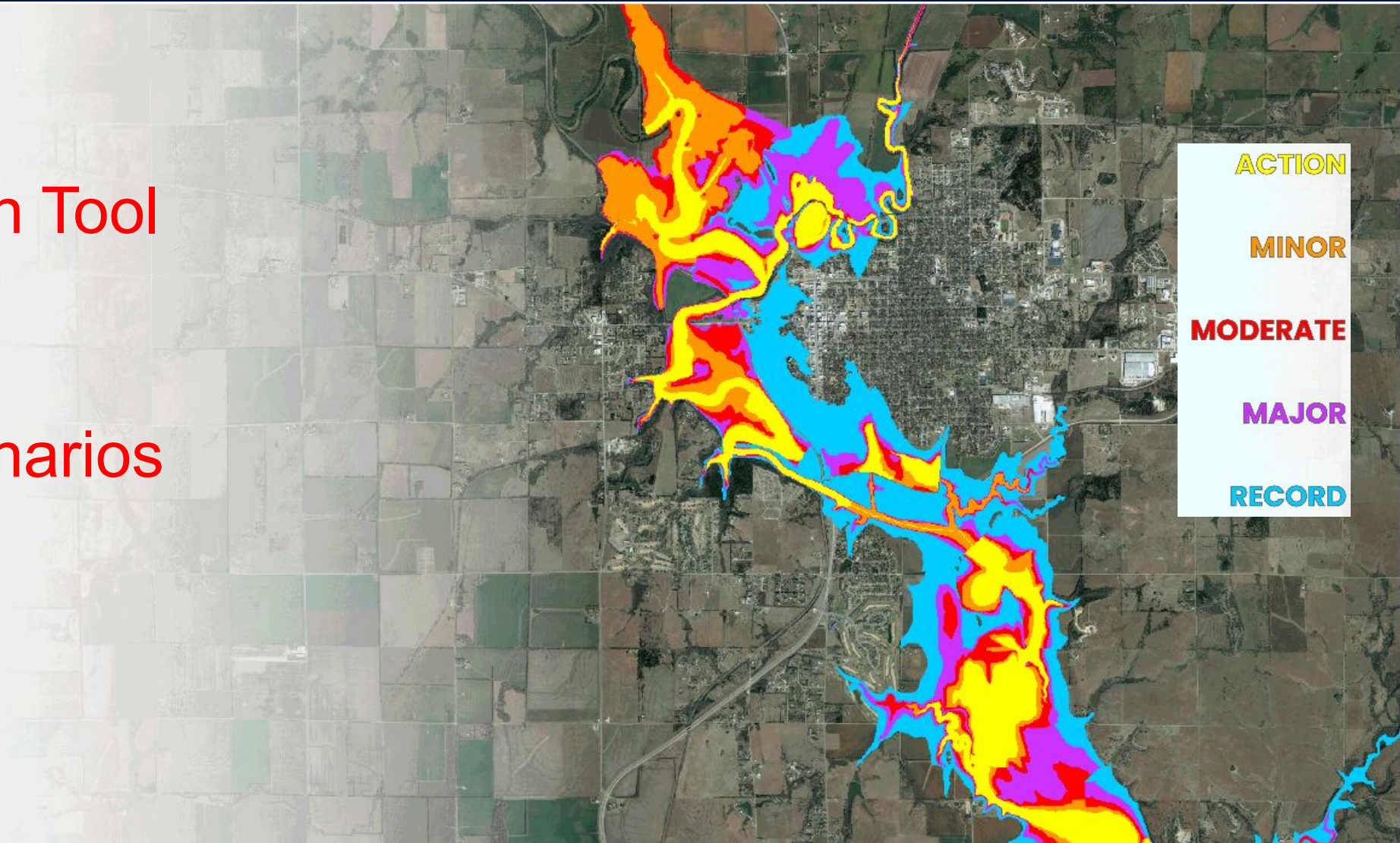
- Static Products
- Up/Downstream of NWPS Gage Locations
- Explicit Flood Categories
 - Action Stage
 - Minor
 - Moderate
 - Major
 - Record
- Two Versions
 - Flow-Based - derived from Synthetic Rating Curves
 - Stage-Based - **derived from Water Surface Elevations**



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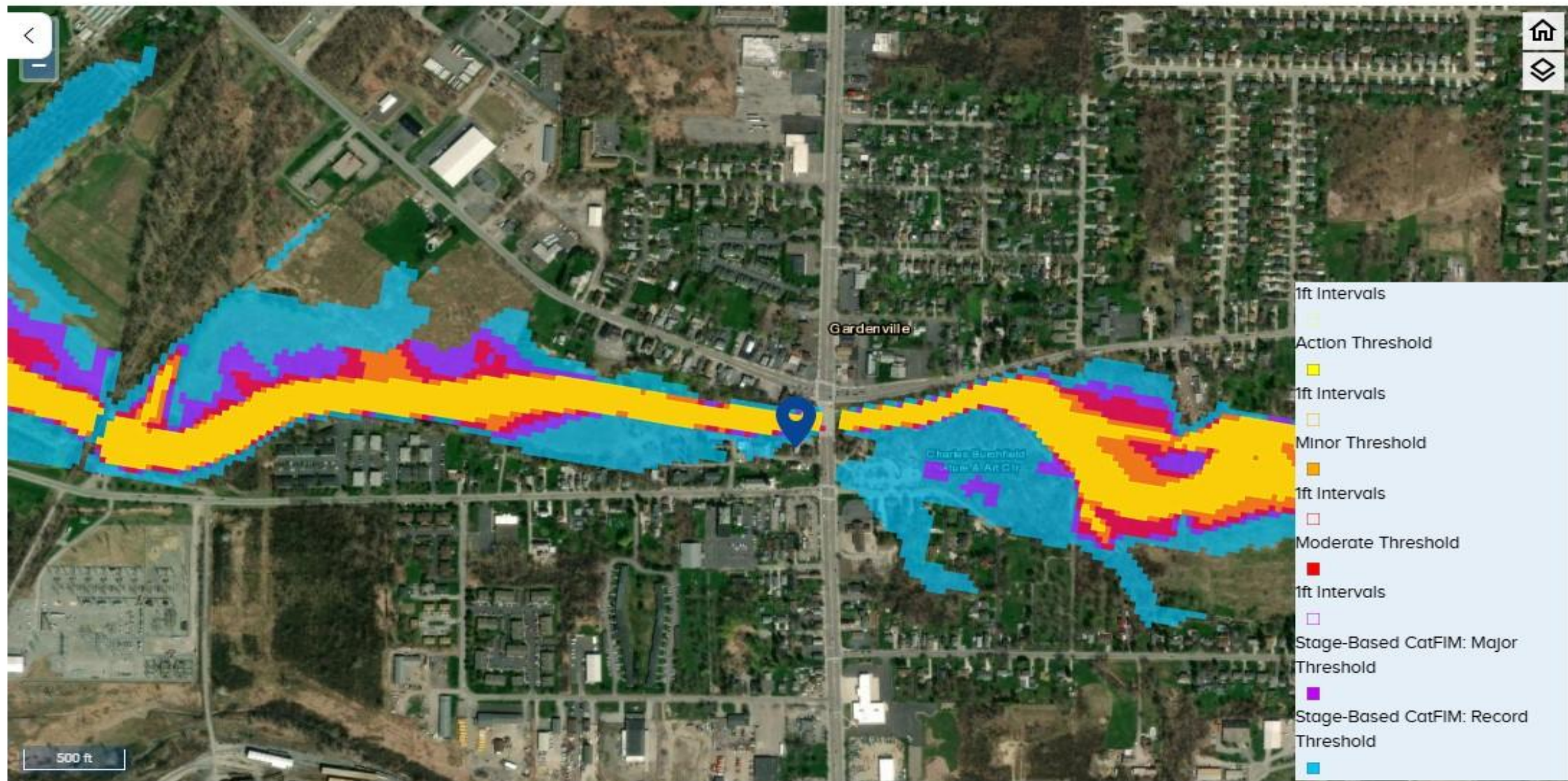
Importance of CatFIMs

- Planning/
Communication Tool
- Quality Control
- “What If?” Scenarios
- Availability



Gauge Location

- ☒ 1ft Intervals
- ☒ Action Threshold
- ☐ 1ft Intervals
- ☒ Minor Threshold
- ☐ 1ft Intervals
- ☒ Moderate Threshold
- ☐ 1ft Intervals
- ☒ Stage-Based CatFIM: Major Threshold
- ☒ Stage-Based CatFIM: Record Threshold

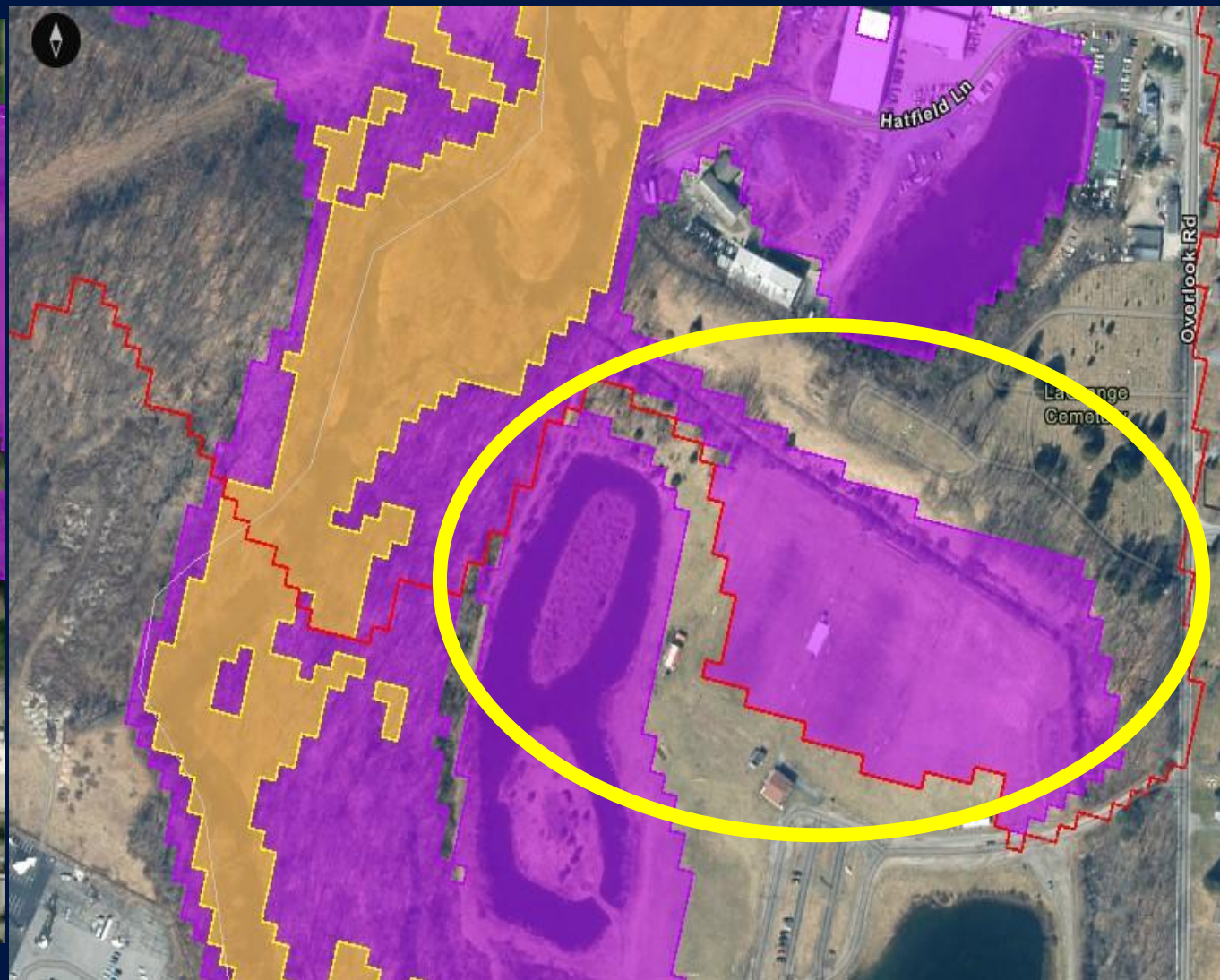


- 1ft Intervals
- Action Threshold
- 1ft Intervals
- Minor Threshold
- 1ft Intervals
- Moderate Threshold
- 1ft Intervals
- Stage-Based CatFIM: Major Threshold
- Stage-Based CatFIM: Record Threshold

- ☒ Display GDVN6 marker
 - ☒ Deactivate GDVN6 Categorical FIM (CatFIM)
- Current opacity: 65%

- ☐ Display FEMA's National Flood Hazard Layers

Reviewing CatFIMs



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Cat FIM Review Coordination

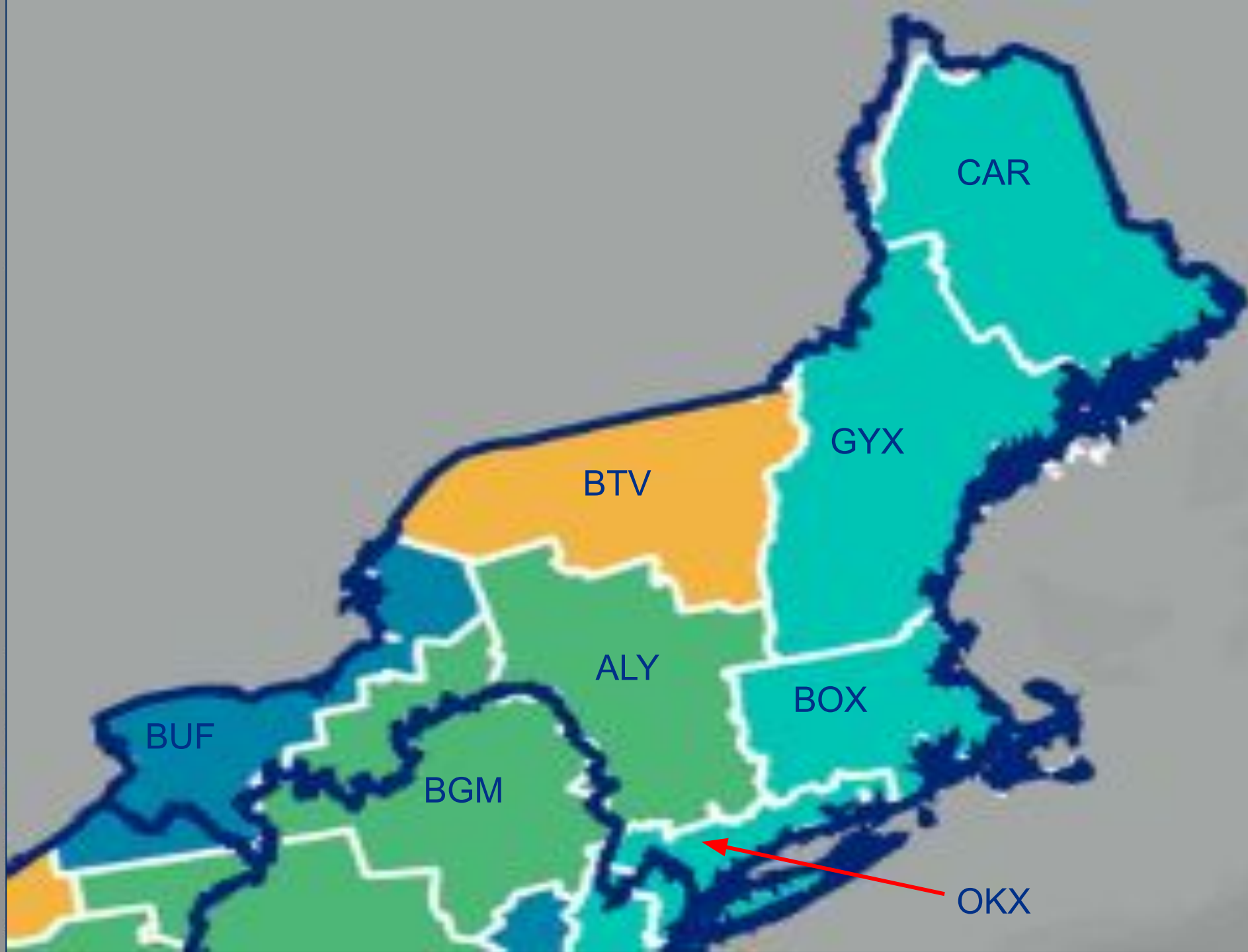
- RFC CatFIM reviews Benefit each of the WFOs
- Recommendations Provided for each Stage CatFIM
 - **Good Overall** - generally good match with Impact Statements (E-19s), limited errors
 - **Use Caution** - issues at some flood thresholds, good match at others
 - **Do Not Use** - significant Over/Underestimation of FIM extents
- WFO SSHs Decide whether to make CatFIM available on NWPS
- All Issues documented with VLab tickets



Map Legend



*100% is approximate. Does not include all parts of Alaska, American Samoa, and Guam. Implementation areas are subject to change.



NERFC CatFIM Review Status - 30% Phase

- 58 Stage CatFIMs for 3 V
 - Good Overall - 22%
 - Use Caution - 40%
 - Do Not Use - 38%
- 55% of Reviewed Stage CatFIMs are Viewable on NWPS
- Many Deficiencies due to Inconsistent Underlying DEM Inconsistencies
- Some Still under Investigation



7 - Moderate flood, residential flooding begins along Route 65 and along Rittenhouse Drive and Creekside Drive in Honeoye Falls and Factory Hollow Road in West Bloomfield. Farmland flooded between Honeoye Falls and Rush.

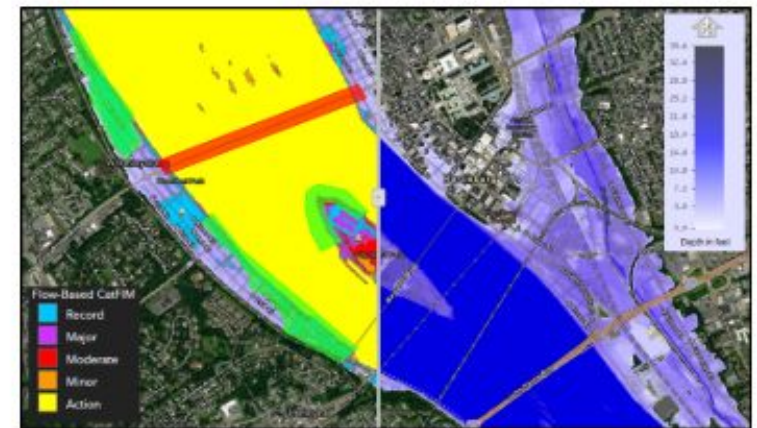


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CatFIM Review Recommendations

- Get All Reviewers on the same page - User's Guide
 - Developed at MARFC
 - Available from NWC
- Stay Ahead of Each Rollout Phase
- Leverage NWC/WFO Reviews of Dynamic FIMs
- Don't Let the "Perfect" be the Enemy of the "Good"

CatFIM Review: A User's Guide



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